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Unit 2 Activity Worksheet | Instruction Formats, Instruction Flow, Addressing Modes & 8086

■ 1. Identifying Instruction Formats

- ADD R1, R2, R3 → Three-address format — the instruction names three operands (two sources R2, R3 and destination R1).
- MOV AX, BX → Two-address format — two operands are specified (source BX, destination AX).
- PUSH AX → One-address format — only one explicit operand (AX); the stack pointer is implied.
- ADD → Zero-address format — no explicit operands; operands are taken from the top of the stack (stack machine style).
- LOAD R1, #10 → Two-address format — a destination register (R1) and a source operand (immediate constant 10) are both specified.

■ 2. Tracing Instruction Flow

- START: MOV AX, #5 → PC points to START; $AX \leftarrow 5$; PC advances to next instruction.
- ADD AX, #3 → $AX \leftarrow 5 + 3 = 8$; PC advances.
- JMP END_LOOP → PC is loaded with the address of END_LOOP (non-sequential jump).
- END_LOOP: HLT → processor halts.
- Note: the instructions at LOOP (INC AX; JMP START) are never executed, because control jumps directly from ADD to END_LOOP; the PC therefore follows the sequence START → (next) → END_LOOP instead of incrementing through every line.

■ 3. Addressing Mode Scenarios

- Loading a constant value directly into a register → Immediate addressing — the operand is part of the instruction, so no memory access is needed and it is fast.
- Accessing an array element using an index stored in another register → Indexed addressing — effective address = base address + index register, ideal for stepping through arrays.
- Retrieving data from a memory location whose address is stored in a register → Register indirect addressing — the register holds the pointer, allowing flexible access to computed locations.
- Jumping to a subroutine whose address is known at assembly time → Direct (absolute) addressing — the target address is encoded directly in the instruction.
- Accessing a variable located at a fixed memory address → Direct addressing — the instruction contains the fixed address of the variable, requiring only one memory reference.

■ 4. 8086 Architecture Diagram (described)

- The 8086 is split into two units. Bus Interface Unit (BIU): contains the segment registers (CS, DS, SS, ES), Instruction Pointer (IP), address-generation logic (physical address = segment $\times$ 16 + offset), the 6-byte instruction queue and the bus control logic — it fetches instructions and data from memory.
- Execution Unit (EU): contains the general-purpose registers (AX, BX, CX, DX, SP, BP, SI, DI), the ALU, the FLAG register and control circuitry — it decodes and executes the queued instructions.

- The two units work in parallel (pipelining): while the EU executes, the BIU prefetches the next instructions into the queue.